

Digital Image Processing

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* DIP: DIP involves converting images into a digital format and performing operations to improve their quality or extract useful information.

→ How it works:

- Input: Digital image (from a camera or scanner).
- Processing: Apply techniques like filtering, sharpening or noise removal.
- Output: A better or analyzed image.

→ Applications:

- Medical imaging (MRI, X-rays).
- Security (facial recognition, CCTV).
- Satellite (weather and land mapping).
- Entertainment (photo editing, special effects).

→ Advantages:

- Enhance image quality.
- Automates analysis tasks.
- Widely used in technology and industries.

* Motivation in DIP

→ DIP is about improving and analyzing pictures using computers. The motivation comes from:

- 1) Improving Picture Quality: Fix blurry, dark, or noisy images to make them clearer and more useful.
- 2) Extracting Information: Identify objects or patterns in pictures, like faces, text or shapes, for tasks like facial recognition or traffic monitoring.
- 3) Wide Use: Used in medicine (X-rays, MRI scans), space, security (CCTV) & entertainment (movies & games).

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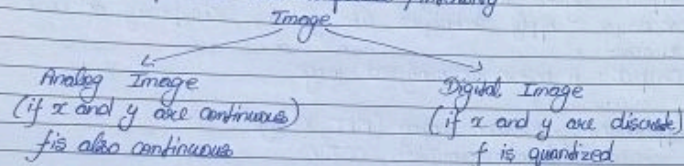
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* Perspective in DIP

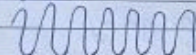
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- 1) Technical view
- 2) Practical uses
- 3) Ethical concerns
- 4) Future Possibilities

$f(x, y)$
where, x, y : represents the spatial coordinates
 f : represents the amplitude/intensity



Analogue Signal: It is a continuous signal that varies smoothly over time.



Digital signal: D's are not continuous; instead, they are sampled at specific intervals.

DIP Advantages

- Improves image quality (brightness, contrast, noise removal).
- Allows image editing and manipulation (resize, crop, filter).
- Useful in many fields (medicine, forensics, remote sensing).
- Enables automated tasks (facial recognition, defect detection).

DIP Disadvantages

- Needs powerful hardware and software.
- Can be expensive (tools and expertise).
- Enables automated tasks (facial recognition, defect detection).
- Risk of misuse (fake or misleading content).

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• Supports data compression for storage and sharing

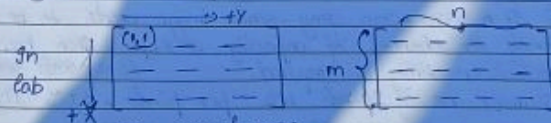
- Loss of detail in compressed images
- Complex techniques may need special skills

* Digital Image Representation

- Monochrome image can be represented as a two-dimensional light intensity function $f(x, y)$
where, x, y are spatial coordinates
the value of f at point (x, y) is the grey level or the brightness of the image

DIP array

	$(0,0)$	$(0,1)$	$(0,2)$
$\downarrow +y$	$(1,0)$	$(1,1)$	$(1,2)$
$+x \rightarrow$	$(2,0)$	$(2,1)$	$(2,2)$



m = no. of rows
 n = no. of columns
Total no. of pixels = $m \times n$
 K = no. of bits/pixels
Total no. of bits = $m \times n \times K$
(Image size)

Numerical

Q. For an image of 512 by 512 pixels, with 8 bits per pixel. Determine the image size?

$m = 512$, $n = 512$, $K = 8$

T.S = $m \times n \times K$

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Q1

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$$\begin{aligned}
 &= 512 \times 512 \times 8 \\
 &= 2097152 \text{ bits} \\
 &\text{Convert into bytes} \quad \{ 1 \text{ byte} = 8 \text{ bits} \} \\
 &= 2097152 / 8 = 262,144 \text{ bytes} \\
 &\text{Convert into Kilo byte} \\
 &= 262,144 / 1024 = 256 \text{ KB} \\
 &\text{Convert into Mega byte} \\
 &= 256 / 1024 = 0.25 \text{ Mb}
 \end{aligned}$$

Q2 What is the storage requirement for a 1024×1024 binary image?

$$\begin{aligned}
 m &= 1024, n = 1024, K = 1 (\text{binary image}) \\
 T.S &= 1024 \times 1024 \times 1 \\
 &= 1048576 \text{ bits} \rightarrow 131,072 \text{ byte} \\
 T.S &= 128 \text{ KB}
 \end{aligned}$$

Q3 If a 375×300 gray-scale image needs to be sent across the channel of capacity 28 Kbps, then how much transmission time is required.

Assuming no. of "pixel" bits per pixel = 8 bits (1 byte) per pixel

$$\begin{aligned}
 m &= 375, n = 300, K = 8 \\
 T.S &= 375 \times 300 \times 8 \\
 &= 900,000 \text{ bits} \rightarrow 112,500 \text{ byte} \\
 &= 109.86 \text{ KB}
 \end{aligned}$$

$$\begin{aligned}
 \text{KBps} &= \text{Kilo bit per sec} \quad [1 \text{ Mbps} = 10^6] \\
 \text{speed} &= 28 \text{ Kbps} = 28 \times 1000 \text{ bps} \\
 \text{time} &= \frac{900,000}{28 \times 1000} = 32.14 \text{ sec}
 \end{aligned}$$

Q4 For an image of 64 by 64 pixels with 3 bands, with 4 bits per pixel. Determine the image size

(Spectral)

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$$\begin{aligned}
 m &= 64, n = 64, K = 4, \text{band} = 3 \\
 T.S &= 64 \times 64 \times 4 \times 3 \\
 &= 49152 \text{ bits} = 6144 \text{ byte} \\
 T.S &= 6 \text{ KB}
 \end{aligned}$$

Q5 For an image of 32 by 32 pixels with 4 bands, in which the smallest unit of an image consists of 11 shades, determine the image size?

$$\begin{aligned}
 m &= 32, n = 32, \text{band} = 4 \\
 \text{11 shades, we check the next integer} \\
 \text{byte 11 in } 2^K, K &= 4 \\
 T.S &= m \times n \times K \times \text{band} \\
 &= 32 \times 32 \times 4 \times 4 \\
 &= 16384 \text{ bits} \\
 &= 2048 \text{ byte} = 2 \text{ KB}
 \end{aligned}$$

K	2^K
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024

Q6 For an image of 25 by 30 pixels with 5 spectral, the pixels have 17 levels, determine the image size?

$$\begin{aligned}
 m &= 25, n = 30, \text{band} = 5, K = 5 \\
 T.S &= 25 \times 30 \times 5 \times 5 \\
 &= 18,750 \text{ bits} = 2343.75 \text{ bytes} \\
 &= 2.34 \text{ KB}
 \end{aligned}$$

Q7 For an image of 1024×1024 pixels with 6 bands (bytes) = 100

$$\begin{aligned}
 T.S &= 1024 \times 1024 \times 6 \times 100 \\
 &= 629,145,600 \text{ bits} = 78600 \text{ KB} \\
 &= 75 \text{ MB}
 \end{aligned}$$

$$T.S = 0.073 \text{ Gb}$$

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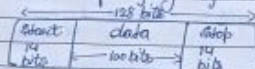
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(b) The size of packet = 188 bit
Start and stop flag of 14 bits each



total no of packets

$$= \frac{1024 \times 1024 \times 6 \times 100}{100} = 1024 \times 1024 \times 6$$

total no of bits to be transmitted

$$= 1024 \times 1024 \times 6 \times 188$$

Speed = 188 x 1000 bps

$$\text{time} = \frac{1024 \times 1024 \times 6 \times 188}{188 \times 1000} = 6291.49 \text{ sec}$$

$$= \frac{6291.49}{3600}$$

$$= 1.74 \text{ hrs}$$

$$[1 \text{ hr} = 3600 \text{ s}]$$

Q10 m = 1280, n = 960, K = 8, Sequence = 200

$$(a) \text{ T.S} = 1280 \times 960 \times 8 \times 200$$

$$= 1966080000 \text{ bits} = 245,760,000 \text{ byte}$$

$$= 240,000 \text{ KB} = 234.3 \text{ GB}$$

(b) Start bit = 14 Stop bit = 14

$$\text{Total bit} = 14 + 14 + 8 \times 1966080000$$

Each packet contain = 8 bit

$$\text{No of packet} = \frac{1966080000}{8} = 245,760,000$$

Total no of bit per packet = 14 + 14 + 8

$$= 36 \text{ bits}$$

total no of bits to be transmitted

$$= 245,760,000 \times 36$$

$$= 8,847,360,000$$

$$\text{speed} = 3 \times 10^6 \text{ bit/sec}$$

$$\{ 1 \text{ baud} = 1 \text{ Mb} = 10^6 \text{ bits/sec} \}$$

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$$\text{time} = \frac{8,847,360,000}{3 \times 10^6} = 2949.12 \text{ sec}$$

(c) Speed = 3×10^6 bits/sec

$$\text{time} = \frac{8,847,360,000}{3 \times 10^6}$$

$$= 2949 \text{ sec}$$

$$\{ 1 \text{ baud} = 1 \text{ Mb} = 10^6 \text{ bits/sec} \}$$

Q11 m = 100, n = 100, K = 8, band = 4

$$\text{T.S} = 100 \times 100 \times 8 \times 4$$

$$= 320000 \text{ bits} = 40,000 \text{ byte}$$

$$= 39.06 \text{ KB}$$

Speed = 10 K bit per second

$$= 10 \times 1000 = 10^4 \text{ bps}$$

$$(i) \text{ time} = \frac{320000}{10^4} = 32 \text{ sec}$$

(ii) m = 100, n = 100, K = 1 (binary image)

$$\text{binary size} = 100 \times 100 \times 1$$

$$= 10000$$

$$\text{No. of binary image} = \frac{320000}{10000} = 32 \text{ images}$$

Q12 m = 100, n = 100, K = 8, band = 4

$$\text{T.S} = 100 \times 100 \times 8 \times 4$$

$$= 320000 \text{ bits}$$

Q13 m = 1024, n = 512

$$\text{No. of bits} = 3 \text{ (RGB)}$$

$$K = 3 \text{ bits}$$

$$\text{T.S} = 1024 \times 1024 \times 3 \times 3$$

$$= 4718592 \text{ bits} = 589,824 \text{ byte}$$

$$= 576 \text{ KB}$$

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Binary Size = $10 \times 10 \times 1$

= 100

no. of binary image = $\frac{4718592}{100} = 47185.92$

Image Sensing: It is in DIP means using a device, like a camera, to capture a picture and turn it into a form that a computer can understand.

ex: A camera takes a photo of something

Electromagnetic Spectrum: It refers to the range of energy waves, like visible light, infrared and x-rays, that can be used to create and process images. Different parts of the spectrum are used depending on the type of image, such as visible light for regular photos or infrared for heat detection.

Components:

- ① Gamma Rays: Highest energy and shortest wavelength.
- Used in Medical imaging (e.g. cancer treatment) & astrophysics.
 - Frequency Range (Hz): $10^{20} - 10^{24}$
 - Wavelength Range: $< 10^{-12}$ m

- ② X-Rays: Used in medical diagnostics (e.g. x-ray imaging) and material analysis.
- Frequency: $10^{16} - 10^{20}$
 - Wavelength: 1 pm - 1 nm

- ③ Ultraviolet (UV): Found in sunlight; used in sterilization, lithography and forensics.
- Frequency: $10^{15} - 10^{17}$
 - Wavelength: 1 nm - 400 nm

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- ④ Visible Light: The range humans can see (e.g. violet to red).
- Used in cameras, displays and general illumination.
 - Freq: $4 \times 10^{14} - 7.5 \times 10^{14}$
 - Wavelength: 400 nm - 750 nm

- ⑤ Infrared (IR): Divided into near, mid and far infrared.
- Used in thermal imaging, remote sensing and communication.
 - Frequency: $10^3 - 10^{14}$
 - Wavelength: 2.5 μ m - 25 μ m

- ⑥ Microwaves: Used in radar, satellite communications, and cooking (microwave ovens).
- Frequency: $3 \times 10^3 - 10^3$
 - Wavelength: 25 μ m - 1 mm

- ⑦ Radio waves: Largest wavelength, lowest frequency.
- Used in communication systems (e.g. radio, TV & mobile phones).
 - Frequency: $< 3 \times 10^3$
 - Wavelength: > 1 mm

* Image Resolution

- It is the ability of the imaging system to produce the smallest discernable details.
- The smallest sized object clearly and differentiable it from the neighbouring small objects that are present in the image.

* Types of Resolution

- Spatial Resolution
- Spectral Resolution
- Radiometric resolution
- Gray level resolution
- Temporal resolution

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① Spatial Resolution : The smallest possible features that can be detected in an image.

OR

- The size of single pixel or area covered by a single pixel (GSD).
- Spatial resolution depends on two parameters:
 - The no. of pixels of the image
 - The no. of bits necessary to adequate intensity resolution referred to as bit depth.
- Line pairs per unit distance
- Dots/pixels per unit distance (dots per inch - dpi)

② Gray Level Resolution : It refers to how many different shades of gray an image can have. It is determined by the bit depth, which is the no. of bits used to represent each pixel's brightness.

- e.g.
- A pixel with an intensity value of 0 represents black.
 - A pixel with an intensity value of 255 represents white.

③ Spectral Resolution : It describes the capability of a sensor to capture information in various bands which have fine wavelength intervals.

- Information recorded by the sensor in various bands are referred as multi-spectral sensors.

e.g. UV-Vis Spectroscopy, FTIR Spectroscopy

④ Temporal Resolution : The concept of revisit period is important for monitoring that help us to identify the changes over a period of time.

- Time taken by a sensor to locate exact same area second time at some interval time is considered as temporal resolution.

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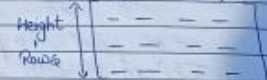
⑤ Radiometric Resolution : It refers to the ability of a sensor or imaging system to measure and distinguish the different levels of intensity of a signal. It determines how accurately the system can measure the brightness of each pixel in an image.

* Image Aspect Ratio

* An aspect ratio is a proportional relationship b/w an image's width and height.

$$\text{Image Aspect Ratio} = \frac{\text{Width}}{\text{Height}}$$

$m \times n$
↓ ↓
rows column



Numerical:

Ques ① If we want to resize a 1024×768 image to one that is 600 pixels wide with the same aspect ratio as the original image, what should be the height of the resized image?

$m = 1024$ - height $n = 768$ - column

$$\text{AR} = \frac{1024}{768} = 1.33$$

$$\text{Aspect} = \frac{600}{H}, \quad 1.33 = \frac{600}{H}, \quad H = 450 \text{ pixels}$$

Ques ② Suppose, we have an image that have 16 rows and have a aspect ratio of 1 or 1:1. Find the total no. of pixels in the image

$$H = 16, \quad \text{Ratio} = 1$$

$$\text{width} = 16 \times 1 = 16$$

$$\text{total no of pixel} = 16 \times 16 = 256$$

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Ques 1) Suppose an image have an aspect ratio of 3:2 with a width of 36 mm. Find the height of the image.

$$\frac{3}{2} = \frac{\text{width}}{\text{height}} = \frac{36}{h}$$

$$\text{height of the image} = \frac{36 \times 2}{3} = 24 \text{ mm}$$

Ques 2) Suppose we have an image that have 80 rows and have an aspect ratio of 2:5. Find the total no of pixel in an image.

$$\frac{2}{5} = \frac{10}{30}, \quad 10 = \frac{40}{5} = 8$$

no of pixel = 8 pixel
Total no of pixel = $8 \times 20 = 160 \text{ pixel}$

Ques 3) Width = no of cols
height = no of rows
Aspect ratio = $\frac{\text{width}}{\text{height}}$

Image I1
rows = 20, column = 10
Aspect ratio = $\frac{10}{20} = 0.5$

Image I2
rows = 100, column = ?, $0.5 = \frac{100}{x}$
A.R = $\frac{10}{11}$, $0.5 = \frac{40}{100}$, $\boxed{10 = 50}$
Total no. of pixels = $100 \times 50 = 5000 \text{ pixel}$
Not exceeding 16 $\rightarrow 0-15$
 $2^K = 16$, $\boxed{K=4}$

Total no of bits = $5000 \times 4 = 20000 \text{ bits}$
Speed = 20 bits/sec

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n bits $\rightarrow$ it represent the data bit, because aggregating is there in packets $\rightarrow$

8 bits		
Start 5 bit	data 10 bits	Stop 5 bit

no. of packets = $\frac{\text{Total size in each packet data}}{10} = \frac{20000}{10} = 2000 \text{ packets}$

Effective no. of bits to be transmitted = $2000 \times 20 = 40,000 \text{ bits}$
time required = $\frac{40,000}{20} = 2,000 \text{ sec}$

* Bit rate: The no. of bits (binary digits) transmitted per second in a communication channel.
Unit: Bits per second (bps)

* Baud Rate: The no. of signal changes transmitted per second in a communication channel. (symbol)
Unit: Baud (or symbols per second).

Bit rate = (Baud rate) $\times$ (no of bit / symbol)
 $R = x \times n$

- R: represents the no. of bits per sec.
- x: represents the no. of symbols per sec or number of elements per sec.
- n: represents the no of bits per symbol or no of bits per element.

Numerical Baud rate

Ques 1 $x = 2400 \text{ baud}$, $n = 1$
 $R = x \times n = 2400 \times 1$

(a) $R = 2400$

(b) Image size = $256 \times 8 = 2048 \text{ bit}$

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Time = $\frac{2048}{2400} = 0.85 \text{ sec}$

Ques 2 $X = 2400, n = 4$
 $R = X \times n = 2400 \times 4$

(a) $R = 9600 \text{ bit}$

(b) Image size = 2048

Time = $\frac{T.S}{R} = \frac{2048}{9600} = 0.21 \text{ sec}$

Ques 3 $X = 640, C = 480, K = 2$

$T.S = 640 \times 480 \times 2$

$T.S = 2457600 \text{ bit}$

(a) $R = X \times n = 9600 \times 1, n = 1, X = 9600$
 $R = 9600$

(b) Time = $\frac{2457600}{9600} = 256 \text{ sec} = 4.26 \text{ min}$

Ques 4 $X = 1920, C = 1080, K = 24$

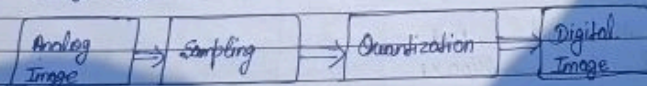
$T.S = 1920 \times 1080 \times 24$

$T.S = 49,766,400$

(a) $R = X \times n = 384,000 \times 4$
 $R = 1536000$

(b) Time = $\frac{49,766,400}{1536000} = 32.4 \text{ sec} = 0.54 \text{ min}$

Image Digitization



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* Sampling: Sampling process converts an analog signal into a discrete signal.

• Image representation by 2D finite matrix

• Digitizing the continuous values is called sampling

* Quantization: Digitizing the amplitude values is called quantization

• It is the process of converting sampled analog pixel intensity into a discrete variable integer

• Each matrix element represented by one of the finite set of discrete values

$$U' \in \{x_1, x_2, x_3, \dots, x_L\}$$

$$U \xrightarrow{\text{Quantization}} U'$$

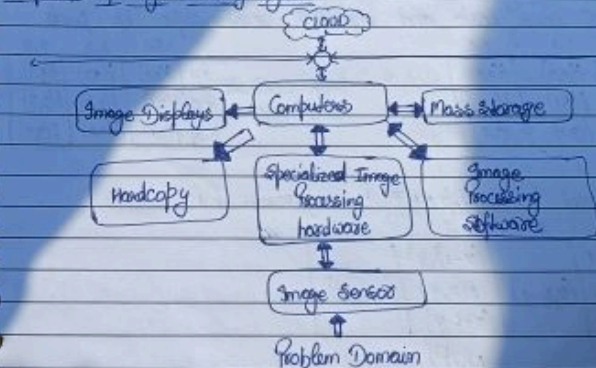
$$U' = x_k \text{ if } \{t_k \leq U < t_{k+1}\}$$

$$t_k, \{k=1, 2, \dots, L+1\}$$

where, t_k : minimum value

t_{L+1} : maximum value

* Components of Image Processing System



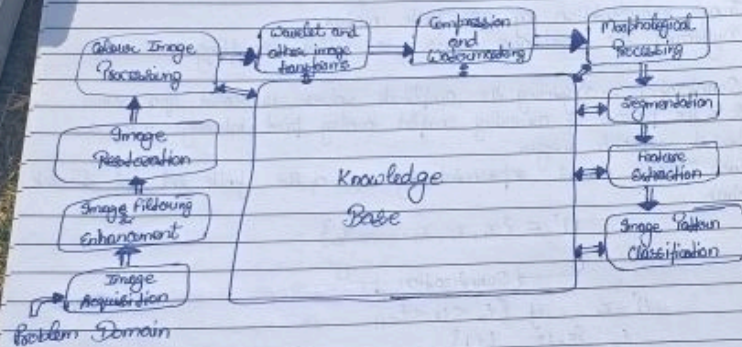
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Fundamental Steps in Image Processing



Relationship Between Pixels

- An image is denoted by a function $f(x, y)$
- Each element $f(x, y)$ at location (x, y) is called a pixel.
- There exist some basic and important relationships b/w pixels.

	$x-2$	$x-1$	x	$x+1$	$x+2$
$y-2$	$f(0,0)$	$f(0,1)$	$f(0,2)$		
$y-1$	$f(1,0)$	$f(1,1)$	$f(1,2)$		
y	$f(2,0)$	$f(2,1)$	$f(2,2)$		
$y+1$					
$y+2$					

Image Coordinates
in MATLAB

	$x-1$	x	$x+1$
$y-1$	$f(1,1)$	$f(1,2)$	$f(1,3)$
y	$f(2,1)$	$f(2,2)$	$f(2,3)$
$y+1$	$f(3,1)$	$f(3,2)$	$f(3,3)$

(i) Neighbourhood Relationship: 4-Neighbours of P

x	$x-1$	x	$x+1$
$y-1$	$(x-1, y-1)$	$P(x, y)$	$(x+1, y-1)$
y	$(x-1, y)$	$P(x, y)$	$(x+1, y)$
$y+1$	$(x-1, y+1)$	$P(x, y)$	$(x+1, y+1)$

$N_4(P)$

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* Diagonal Neighbour

$x-1, y-1$	x, y	$x+1, y+1$
$(x-1, y-1)$	$P(x, y)$	$(x+1, y+1)$
$(x+1, y-1)$		

* 8-Neighbour of P

$x-1, y-1$	$x-1, y$	$x-1, y+1$
$(x-1, y-1)$	$P(x, y)$	$(x, y+1)$
$(x+1, y-1)$	$(x+1, y)$	$(x+1, y+1)$

(ii) Adjacency

- 4-adjacency: $P, q \in \{v\}$
 $q \in N_4(P)$

*	*	*
*	P	*
*	*	*

$N_4(P)$

- 8-adjacency: $P, q \in \{v\}$
 $q \in N_8(P)$

*	*	*
*	P	*
*	*	*

- mixed-adjacency: $P, q \in \{v\}$

$$(a) q \in N_4(P) \quad \text{OR}$$

$$(b) q \in N_8(P)$$

$$N_4(P) \cap N_8(P) = \{P\}$$

Ques 1) Compute the length of shortest-4 path b/w pixels P & q, where $v = \{1, 2\}$

	0	1	2	3
0	4	2	3	2(q)
1	3	3	1	3
2	2	2	3	2
3	2(P)	1	2	3

Shortest 4-path: No shortest 4-path (because from (1,2) to (0,3) No path exist)

Ques 2) Shortest 8-path:

$$2(P) - 1 - 2 - 1 - 2(q)$$

$$\text{Length} = 4$$

	0	1	2	3
0	4	2	3	2(q)
1	3	3	1	3
2	2	2	3	2
3	2(P)	1	2	3

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Ques 3) m-adjacency

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4	2	3	2(a)
2	2	1	3
2	3	2	2
3(p)	1	2	3

Ques 4) $V = \{a, b\}$

Shortest 4-path: no path

3	1	2	1(a)
2	2	0	2
1	2	1	1
1(p)	0	1	2

Shortest 3-path: $1(p) \rightarrow a \rightarrow 1 \rightarrow 1(a)$
length = 4

2	1	2	1(a)
2	2	0	2
1	2	1	1
1(p)	0	1	2

m-adjacency:

3	1	2	1(a)
2	2	0	2
1	2	1	1
1(p)	0	1	2

Ques 5) $V = \{1, 2\}$

Shortest 4-path: $1(p) \rightarrow 1 \rightarrow 2 \rightarrow 1 \rightarrow 1(a)$

length = 6, $1(p) \rightarrow 1 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow 1(a)$
 $1(p) \rightarrow 2 \rightarrow 1 \rightarrow 1 \rightarrow 2 \rightarrow 1(a)$

Shortest 2-path

$1(p) \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow 1(a)$ length = 5

2	1	2	1(a)
2	2	0	2
1	2	1	1
1(p)	0	1	2

$1(p) \rightarrow 2 \rightarrow 2 \rightarrow 1 \rightarrow 1(a)$ length = 4

2	1	2	1(a)
2	2	0	2
1	2	1	1
1(p)	0	1	2

m-adjacency: $N_4(p) = \{b\}$

$N_4(i) = \{h, e, j\}$

$N_4(p) \cap N_4(i) = \{b\} \cap \{h, e, j\} = \emptyset$

2 ^a	1 ^b	2 ^c	1 ^d
2 ^e	2 ^f	0 ^g	2 ^h
1 ⁱ	2 ^j	1 ^k	1 ^l
1 ^m	0 ⁿ	1 ^o	2 ^p

Ques 6) Consider the following neighbours of the pixel $(1,1)$, $V = \{1, 2, 3\}$

$\Rightarrow$ 4-adj

$b(c,1)$

$d(1,0)$

$e(1,2)$

0	2	1	0	2
1	3	d	3	e
2	0	f	5	g

$\Rightarrow$ 8-adj: $N_8(p) = \{b, d, e\}$

$N_8(i) = \{h, e, j\}$

$N_8(p) \cap N_8(i) = \{b\} \cap \{h, e, j\} = \emptyset$ not

$N_4(p) = \{b, d, e\}$

$N_4(c) = \{b, e\}$

$N_4(p) \cap N_4(c) = \{b, e\} \cap \{b, e\} = \{b, e\}$

$p \times c$ are not adj

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$$N_1(p) = \{a, b, c\} \\ N_1(h) = \{c\} \quad , \quad N_1(p) \cap N_1(h) = \{c\} = \{c\}$$

Ans $\begin{matrix} h(a,1) \\ d(1,2) \\ e(1,2) \end{matrix}$

Region

- For two image subsets: S_i and S_j are adjacent if $p \in S_i$ and $q \in S_j$ such that p and q are adjacent

	R_i	← Independent pixel →	R_j	
1	10	01	00	10
1	00	10	00	10
1	10	00	10	10

ex $V = \{1, 2\}$

- 4-adj $\Rightarrow R_i$ & R_j are not 4-adjacent
Since, p & q are not 4-adj
- 8-adj $\Rightarrow R_i$ & R_j are 8-adj
So, p & q are also 8-adj
- m-adj $\Rightarrow N_1(p) = \{1, 2\}$
 $N_1(q) = \{2\}$
 $N_1(p) \cap N_1(q) = \{2\}$
So, p & q are m-adjacent

Ques (1) S_1 and S_2 , $V = \{1, 2\}$

- 4-adj $\Rightarrow p$ & q are not 4-adj
So, S_1 & S_2 are not 4-adj
- 8-adj $\Rightarrow p$ & q are diagonal to each other
So, S_1 & S_2 are 8-adj

	S_1					S_2			
0	0	0	0	0	0	1	1		
0	0	10	01	00	10	0	0		
0	0	10	01	10	10	0	0		
0	1	10	10	00	00	0	0		

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m-adj: $\begin{matrix} m(p) = \{1, 2\} \\ m(q) = \{2\} \\ m(p) \cap m(q) = \{2\} \\ p \text{ \& } q \text{ are m-adj} \\ \text{So, } S_1 \text{ \& } S_2 \text{ are m-adj} \end{matrix}$

Distance Measure

- Euclidean Distance: $D_e(p, q) = [(u-w)^2 + (v-x)^2]^{1/2}$
- City Block Distance: $D_u(p, q) = |u-w| + |v-x|$
- Cross Street Distance: $D_c(p, q) = \max(|u-w|, |v-x|)$

$p(u, v) \quad q(w, x)$

ex

	0	1	2	3	4
0					
1					
2			(2,2)		
3					
4					

from (2,2) we find the distance of all pixel

Ques (2) Let $V = \{1, 2\}$ and D_e, D_u and D_c distance b/w two pixel p & q

	0	1	2	3
0	0	1	1	1
1	1	0	0	1
2	1	1	1	1(2)
3	1(1)	1	1	1

$\begin{matrix} (2,0) \\ (4,0) \end{matrix}$

$D_e(p, q) = \sqrt{(3-2)^2 + (0-3)^2}$
 $= \sqrt{1+9} = \sqrt{10}$

$D_u(p, q) = |3-2| + |0-3|$
 $= 1+3 = 4$

$D_c(p, q) = \max(|3-2|, |0-3|)$
 $= \max(1, 3) = 3$

$D_s(p, q) = 3 \text{ unit}$

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Ques 13) $f(x,y)$ and $g(x,y)$ and distance b/w pixels p & q

$f(x,y)$ $g(x,y)$
 $p(5,0)$ $q(4,4)$
 $D_e(p,q) = \sqrt{(5-4)^2 + (0-4)^2} = \sqrt{16+16} = \sqrt{32}$
 $D_c(p,q) = |5-4| + |0-4| = 4+4 = 8$
 $D_m(p,q) = 4 \max$

Image Arithmetic Operation

- Image Addition: $g(x,y) = f_1(x,y) + f_2(x,y)$
- Image Subtraction: $g(x,y) = f_1(x,y) - f_2(x,y)$
- Image Multiplication: $g(x,y) = f_1(x,y) \times f_2(x,y)$
- Image Division: $g(x,y) = \frac{f_1(x,y)}{f_2(x,y)}$
- Image Blending: $g(x,y) = \alpha \times f_1(x,y) + (1-\alpha) \times f_2(x,y)$
 α = Blending ratio, $\alpha = 0.6$ given

Ques 14)

f_1	1	3	7
	5	15	35
	300	50	150

f_2	50	150	125
	45	55	155
	200	50	75

$f_1 + f_2$	51	153	132
	50	70	230
	400	100	225

$f_1 \times f_2$	51	153	132
	50	70	230
	255	100	225

unit 8 range is
 0 to 255
 0 to 255
 Result > 255
 O/P = 255
 Result < 0
 O/P = 0
 0 ≤ Result ≤ 255
 No change

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$f_1 - f_2$	-49	-142	-118
	-40	-40	-80
	0	0	75

$f_1 \times f_2$	50	450	875
	225	225	1125
	4000	2000	11250

f_1 / f_2	0.02	0.02	0.06
	0.11	0.27	0.42
	1	1	2

For Blending operation, Assume $\alpha = 0.6$

$\alpha \times f_1$	1	2	4
	3	9	45
	180	30	90

$(1-\alpha) \times f_2$	30	40	50
	13	22	62
	20	20	30

$\alpha \times f_1 + (1-\alpha) \times f_2$	31	33	54
	21	31	107
	200	50	120

Intensity Transformation

- Point transformation
- Linear transformation

Q 15)

1	2	3
0	1	1
2	2	3

1. greatest no = 3
 next 1 integer no = 4
 $L = 4$
 $S = L - 1 - X$
 $S = 4 - 1 - X$, $S = 3 - X$

O/P image

2	1	0
3	2	2
1	1	0

Q 16) $L = 1024$
 $S = L - 1 - X = 1023 - X$

O/P image

133	818	918
832	771	924
322	373	947

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- Non-linear transformation : $S = C \log(1+x)$
- Power law transformation : $S = Cx^Y$

Ques 5)

0	2	1	1	2
2	3	4	5	2
6	7	6	7	2
2	6	5	6	1
1	0	3	3	1

$n=3$
 $L=2^n=8$
 $A=3, B=5$

without Background

$3 \leq x \leq 5$

$c/p = L-1$

$= 7$

else

$c/p = 0$

with Background

$3 \leq x \leq 5$

$c/p = 7$

else

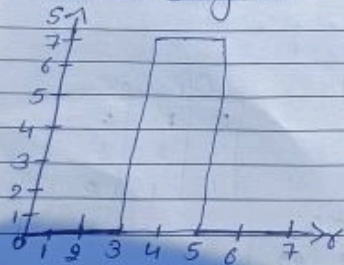
$c/p = \text{same}(x/p)$

0	0	0	0	0
0	7	7	7	2
0	0	0	0	0
0	0	7	0	0
0	0	7	7	0

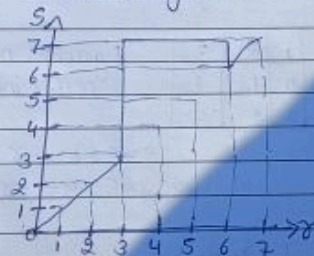
2	2	1	1	2
2	7	7	7	2
6	7	6	7	2
2	6	7	6	1
1	0	7	7	1

Graph:

without Background



with background



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Bit Plane Slicing

$n=8$
 0 to 255
 $(7)_{10} \rightarrow 00000000$ (LSB)
 bit plane (holds most information)
 bit plane 0 (least information) or fine details

Ques 6)

4	9	1	0
1	8	5	7
5	1	2	15
2	4	6	7

$\Rightarrow$ 4bit $\Rightarrow$

0102	01001	0001	0000
0008	0010	0101	0111
0105	0001	0010	1111
0012	0100	0110	0111

a) 0th bit plane

0	1	1	0
1	0	1	1
1	1	0	1
0	0	0	1

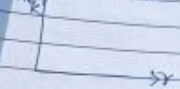
- (*) if value is greater than or equal to 3, so value is 1 otherwise 0 (less than)

1	1	0	0
0	0	1	1
1	0	0	1
0	1	1	1

Image Enhancement (Histogram)

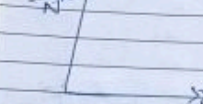
Frequency of occurrence of each gray level

1) Un-normalised



x -axis: gray level
 y -axis: no. of occurrence of each gray level

2) Normalised



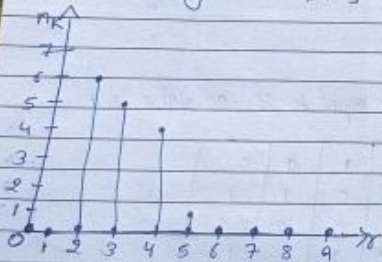
x -axis: gray level
 y -axis: probability of occurrence of each gray level

Ques 1) (a) Un-normalised histogram

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

Gray scale = [0, 9]

x_k	n_k
0	0
1	0
2	6
3	5
4	4
5	0
6	0
7	0
8	0
9	0



• agar ques me sirf histogram hai to by default un-normalised draw karna hai

(b) Normalised

$\frac{n_k}{N}$



x_k	n_k	$\frac{n_k}{N}$
0	0	0
1	0	0
2	6	0.38
3	5	0.31
4	4	0.25
5	1	0.06
6	0	0
7	0	0
8	0	0
9	0	0

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Ques 2

gray scale not mention
 So, take Maximum no = 6
 $L=2 \Rightarrow x$ -axis $\rightarrow$ next integer x^k
 [0, 7]

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Histogram Equalisation : (HE)

Ques 1

r_k	n_k	n_k/N	$CDF(E)$	$(L-1)E$	O/P gray level s_k
0	790	0.19	0.19	1.93	1
1	1023	0.25	0.44	3.08	2
2	850	0.21	0.65	4.55	5
3	656	0.16	0.81	5.67	6
4	329	0.08	0.89	6.23	6
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1.00	7.00	7

$N/\Sigma = 4096$ $L = 8$ (given)
 Σ Total no of pixel $L-1 = 8-1 = 7$

equivalent s_k	n_k	n_k/N
0	0	0
1	790	0.19
2	0	0
3	1023	0.25
4	0	0
5	850	0.21
6	$656 + 329 = 985$	0.24
7	$245 + 122 + 81 = 448$	0.11

Total = 4096

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sign = 5, no. of bit = 16

Ques 2 Perform Histogram Equalisation

1	2	3	4
5	5	6	6
6	7	6	6
6	7	2	3

$L = 8$
 $r_k = 0$ to $L-1$
 0 to 7

r_k	n_k	n_k/N	$CDF(E)$	$(L-1)E$	O/P gray level s_k
0	0	0	0	0	0
1	1	0.06	0.06	0.48	0
2	2	0.13	0.19	1.33	1
3	2	0.13	0.32	2.24	2
4	1	0.06	0.38	2.66	3
5	2	0.13	0.51	3.57	4
6	6	0.38	0.89	6.23	6
7	2	0.13	1.02	7.14	7

$\Sigma = 16$

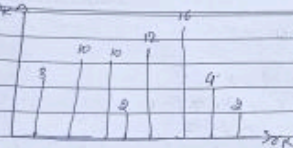
after equalisation

construct the histogram
 $\rightarrow$ o/p image

equalisation s_k	n_k	n_k/N	0	1	2	3
0	1	0.06	4	4	6	6
1	2	0.13	6	7	6	6
2	2	0.13	6	7	1	2
3	1	0.06				
4	2	0.13				
5	0	0				
6	26	0.38				
7	22	0.13				

$\Sigma = 16$

Ques 4)



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Histogram Matching

↳ Input image as well as the desired histogram pattern is also defined

Ques Perform Histogram matching for the following 8x8, eight level image described below

Table 1: Pixel distribution of image

x_k	0	1	2	3	4	5	6	7
n_k	8	10	10	2	12	16	4	2

Table 2: Target histogram

x_k	0	1	2	3	4	5	6	7
n_k	0	0	0	0	20	20	16	8

Step 1: perform H.E on Table 1

$L=8, L-1=7$

x_k	n_k	n_k/N	$(Cpdf)_E$	$(L-1)\Sigma$	S_k
0	8	0.13	0.13	0.91	1
1	10	0.16	0.29	2.03	2
2	10	0.16	0.45	3.15	3
3	2	0.03	0.48	3.36	3
4	12	0.19	0.67	4.69	5
5	16	0.25	0.92	6.44	6
6	4	0.06	0.98	6.86	7
7	2	0.03	1.01	7.07	7

$\Sigma=64$

Step 2: Perform H.E on Table 2



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$$x_{\min} = 6, \quad x_{\max} = 15$$

$$s_{\min} = 0, \quad s_{\max} = 7$$

$$S = \left[\frac{x-6}{15-6} \right] (7-0) + 0$$

$$S = \left[\frac{x-6}{9} \right] (7) \quad 6 \leq x \leq 15$$

$$S = \left[\frac{x-6}{9} \right] (7)$$

$$S = \left[\frac{x-6}{9} \right] (7)$$

x	S	round off
6	$\frac{7}{9} \times 0$	0
7	$\frac{7}{9} \times (7-6)$	1
8	$\frac{7}{9} \times (8-6)$	2
9	$\frac{7}{9} \times (9-6)$	3
10	$\frac{7}{9} \times (10-6)$	4
11	$\frac{7}{9} \times (11-6)$	5
12	$\frac{7}{9} \times (12-6)$	5
13	$\frac{7}{9} \times (13-6)$	6
14	$\frac{7}{9} \times (14-6)$	6
15	$\frac{7}{9} \times (15-6)$	7

Type (2)

Example (2)

$$x_{\min} = 2, \quad x_{\max} = 5$$

$$s_{\min} = 5, \quad s_{\max} = 14$$

$$S = \left[\frac{x-2}{5-2} \right] (14-5) + 5$$

$$S = \left[\frac{x-2}{3} \right] (9) + 5 \quad 2 \leq x \leq 5$$

$$S = \left[\frac{x-2}{3} \right] (9) + 5 = 0 + 5$$

$$S = \left[\frac{x-2}{3} \right] (9) + 5 = 1 + 5$$

$$S = \left[\frac{x-2}{3} \right] (9) + 5 = 2 + 5 = 7 \quad (x=4)$$

$$S = \left[\frac{x-2}{3} \right] (9) + 5$$

$$S = \left[\frac{x-2}{3} \right] (9) + 5 \quad (x=2)$$

$$= 3x - x + 5 = 2x + 5$$

$$x=2, S=5 \quad x=4, S=11$$

$$x=3, S=8 \quad x=5, S=14$$

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x	S	x	S	x	S	x	S	x	S
2	11	11	11	11	11	11	11	11	11
3	11	5	5	5	5	11	11	11	11
4	11	5	14	14	5	11	11	11	11
5	11	5	14	14	5	11	11	11	11
6	11	5	5	5	5	11	11	11	11
7	11	11	11	11	11	11	11	11	11
8	8	8	8	8	8	8	8	8	8

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